

# Doctors Hub - Comprehensive Data Structure Report

[!NOTE]

This document details the complete data structure of the **Doctors Hub** platform, encompassing Django backend models, database schema relations, junction entity patterns, REST API serializers, and frontend state contracts.

## 1. Executive Summary & Architectural Overview

The **Doctors Hub** data architecture follows a normalized relational structure powered by Django ORM with UUID primary keys across all models. The domain is divided into five core business modules:

- Authentication & User Management:** Custom phone-number-based user authentication.
- Medical Practitioner Domain:** Doctor profiles, specialties, affiliations across facilities, schedules, and chamber appointments.
- Hospital Facilities Domain:** In-patient and OPD Multi-Specialty institutes, categories, services, internal diagnostics, and affiliated doctors.
- Diagnostic Center & Lab Domain:** Standalone diagnostic centers, center classifications, diagnostic services, master lab test catalog, and center-specific test pricing.
- Patient Bookings:** Doctor chamber bookings and doorstep / lab test sample collection bookings.

[Entity Relationship Overview]

User → DoctorBooking / LabBooking  
Doctor → DoctorSpecialty / DoctorAffiliation  
Hospital / DiagnosticCenter → DoctorAffiliation  
Hospital / DiagnosticCenter → DiagnosticCenterTest  
Test → DiagnosticCenterTest

## 2. Detailed Data Schema & Model Definitions

### 2.1 Authentication & User Domain

User

Replaces Django's default user model with `phone_number` as the primary identifier for patient authentication.

| Field                     | Type                       | Attributes / Constraints                               | Description               |
|---------------------------|----------------------------|--------------------------------------------------------|---------------------------|
| <code>id</code>           | <code>UUIDField</code>     | Primary Key, <code>default=uuid4</code> , Non-editable | Unique User ID            |
| <code>phone_number</code> | <code>CharField(15)</code> | <code>unique=True</code>                               | Unique login phone number |
| <code>first_name</code>   | <code>CharField(50)</code> | <code>blank=True</code>                                | User given name           |
| <code>last_name</code>    | <code>CharField(50)</code> | <code>blank=True</code>                                | User surname              |
| <code>is_active</code>    | <code>BooleanField</code>  | <code>default=True</code>                              | Account status flag       |
| <code>is_staff</code>     | <code>BooleanField</code>  | <code>default=False</code>                             | Admin access flag         |
| <code>date_joined</code>  | <code>DateTimeField</code> | <code>default=timezone.now</code>                      | Registration timestamp    |

2.2 Medical Practitioner Domain

DoctorSpecialty

Master catalog of medical specialties (e.g., Cardiology, Neurology, Pediatrics).

| Field       | Type           | Constraints           | Description            |
|-------------|----------------|-----------------------|------------------------|
| id          | UUIDField      | PK, default=uuid4     | Specialty Identifier   |
| name        | CharField(100) | Required              | Name of specialty      |
| slug        | SlugField(120) | unique=True           | URL slug               |
| icon        | CharField(50)  | default='Stethoscope' | Lucide icon identifier |
| description | TextField      | blank=True            | Detailed overview      |

Doctor

Physician profile containing qualification details and specialty mappings.

| Field         | Type            | Constraints                | Description                        |
|---------------|-----------------|----------------------------|------------------------------------|
| id            | UUIDField       | PK, default=uuid4          | Doctor Identifier                  |
| name          | CharField(200)  | Required                   | Full doctor name with titles       |
| specialties   | ManyToManyField | References DoctorSpecialty | Associated specialties             |
| qualification | TextField       | Required                   | Medical degrees (e.g., MBBS, FCPS) |
| experience    | CharField(50)   | Required                   | Years / description of experience  |

DoctorAffiliation (Key Polymorphic Junction)

Connects a Doctor to either a Hospital OR a DiagnosticCenter along with consultation fees and type.

| Field             | Type              | Constraints                                  | Description                       |
|-------------------|-------------------|----------------------------------------------|-----------------------------------|
| id                | UUIDField         | PK, default=uuid4                            | Affiliation Identifier            |
| doctor            | ForeignKey        | FK Doctor , on_delete=CASCADE                | Associated doctor                 |
| hospital          | ForeignKey        | FK Hospital , null=True , blank=True         | Hospital facility (if applicable) |
| diagnostic_center | ForeignKey        | FK DiagnosticCenter , null=True , blank=True | Diagnostic center (if applicable) |
| consultation_type | CharField(50)     | Choices: ['OPD', 'In-patient']               | Type of consultation              |
| fee               | DecimalField(8,2) | Required                                     | Consultation fee in BDT           |

AffiliationSchedule

Defines available days and visiting time slots for a specific DoctorAffiliation .

| Field       | Type       | Constraints                              | Description            |
|-------------|------------|------------------------------------------|------------------------|
| id          | UUIDField  | PK, default=uuid4                        | Schedule Identifier    |
| affiliation | ForeignKey | FK DoctorAffiliation , on_delete=CASCADE | Associated affiliation |

| Field       | Type          | Constraints                 | Description         |
|-------------|---------------|-----------------------------|---------------------|
| day_of_week | CharField(20) | e.g. 'Saturday' , 'Mon-Wed' | Day(s) available    |
| start_time  | TimeField     | Required                    | Visiting start time |
| end_time    | TimeField     | Required                    | Visiting end time   |

## 2.3 Hospital Facilities Domain

### HospitalCategory

Classification categories for hospitals (e.g., Multi-Specialty, Cardiac Institute).

| Field       | Type           | Description               |
|-------------|----------------|---------------------------|
| id          | UUIDField      | PK                        |
| name        | CharField(100) | Category name             |
| slug        | SlugField(120) | URL slug                  |
| icon        | CharField(50)  | Icon identifier           |
| description | TextField      | Description               |
| count       | IntegerField   | Registered hospital count |

### HospitalService

Specific emergency and inpatient services (e.g., 24/7 ICU, Emergency, CCU).

| Field       | Type           | Description     |
|-------------|----------------|-----------------|
| id          | UUIDField      | PK              |
| name        | CharField(150) | Service name    |
| icon        | CharField(50)  | Icon identifier |
| description | TextField      | Service details |

### Hospital

Primary institute model for hospitals, containing location metadata and service relationships.

| Field      | Type            | Constraints      | Description              |
|------------|-----------------|------------------|--------------------------|
| id         | UUIDField       | PK               | Unique Hospital ID       |
| name       | CharField(200)  | Required         | Institute name           |
| branch     | CharField(200)  | blank=True       | Branch / location suffix |
| slug       | SlugField(220)  | unique=True      | Auto-generated slug      |
| categories | ManyToManyField | HospitalCategory | Categories               |
| services   | ManyToManyField | HospitalService  | Provided services        |
| address    | CharField(300)  | blank=True       | Full street address      |

| Field                 | Type           | Constraints   | Description                |
|-----------------------|----------------|---------------|----------------------------|
| district              | CharField(100) | blank=True    | District name              |
| division              | CharField(100) | blank=True    | Division name              |
| city                  | CharField(100) | blank=True    | City name                  |
| phone                 | CharField(50)  | blank=True    | Contact phone              |
| email                 | EmailField     | blank=True    | Official email             |
| rating                | FloatField     | default=0.0   | Rating score               |
| reviews_count         | IntegerField   | default=0     | Review count               |
| open_timing           | CharField(100) | blank=True    | Operational hours          |
| has_diagnostic_center | BooleanField   | default=True  | Internal lab facility flag |
| is_verified           | BooleanField   | default=False | Verification badge flag    |

## 2.4 Diagnostic Center & Laboratory Test Domain

### TestCategory

Hierarchical category tree for laboratory and pathology tests (e.g., Biochemistry, Cardiac Tests, USG, MRI).

| Field | Type                 | Description   |
|-------|----------------------|---------------|
| id    | UUIDField            | PK            |
| name  | CharField(150)       | Category name |
| slug  | SlugField(170)       | URL slug      |
| icon  | CharField(100)       | Category icon |
| order | PositiveIntegerField | Sort order    |

### Test

Master catalog of standardized medical tests.

| Field                    | Type                 | Description                                     |
|--------------------------|----------------------|-------------------------------------------------|
| id                       | UUIDField            | PK                                              |
| category                 | ForeignKey           | FK TestCategory                                 |
| name                     | CharField(200)       | Test name (e.g., "MRI - ANKLE JOINT BILATERAL") |
| slug                     | SlugField(220)       | Auto-generated slug                             |
| code                     | CharField(50)        | Test code                                       |
| sample_type              | CharField(100)       | e.g. Blood, Urine, Tissue                       |
| preparation_instructions | TextField            | Fasting / prep instructions                     |
| fasting_required         | BooleanField         | Fasting flag                                    |
| report_time_hours        | PositiveIntegerField | Turnaround time in hours                        |

### DiagnosticCenterCategory

Classification tree for diagnostic centers (e.g., Pathology, Imaging, Government, Corporate Chain). Supports self-referencing `parent` hierarchy.

| Field               | Type                        | Constraints                                               | Description     |
|---------------------|-----------------------------|-----------------------------------------------------------|-----------------|
| <code>id</code>     | <code>UUIDField</code>      | PK                                                        | Category ID     |
| <code>name</code>   | <code>CharField(150)</code> | Required                                                  | Category name   |
| <code>slug</code>   | <code>SlugField(170)</code> | <code>unique=True</code>                                  | Category slug   |
| <code>parent</code> | <code>ForeignKey</code>     | Self-referencing FK <code>DiagnosticCenterCategory</code> | Parent category |

### DiagnosticCenter

Standalone diagnostic facility entity.

| Field                   | Type                         | Description                                                 |
|-------------------------|------------------------------|-------------------------------------------------------------|
| <code>id</code>         | <code>UUIDField</code>       | PK                                                          |
| <code>name</code>       | <code>CharField(250)</code>  | Facility name                                               |
| <code>branch</code>     | <code>CharField(200)</code>  | Branch name                                                 |
| <code>categories</code> | <code>ManyToManyField</code> | <code>DiagnosticCenterCategory</code>                       |
| <code>address</code>    | <code>CharField(300)</code>  | Address                                                     |
| <code>district</code>   | <code>CharField(100)</code>  | District                                                    |
| <code>division</code>   | <code>CharField(100)</code>  | Division                                                    |
| <code>tests</code>      | <code>ManyToManyField</code> | <code>Test</code> through <code>DiagnosticCenterTest</code> |

### DiagnosticCenterTest (Key Through Table)

Links a `Test` to either a `DiagnosticCenter` OR a `Hospital`, overriding prices, discounts, and turnaround times per facility.

| Field                               | Type                            | Description                                                                         |
|-------------------------------------|---------------------------------|-------------------------------------------------------------------------------------|
| <code>id</code>                     | <code>UUIDField</code>          | PK                                                                                  |
| <code>center</code>                 | <code>ForeignKey</code>         | FK <code>DiagnosticCenter</code> , <code>null=True</code> , <code>blank=True</code> |
| <code>hospital</code>               | <code>ForeignKey</code>         | FK <code>Hospital</code> , <code>null=True</code> , <code>blank=True</code>         |
| <code>test</code>                   | <code>ForeignKey</code>         | FK <code>Test</code> , <code>on_delete=CASCADE</code>                               |
| <code>price</code>                  | <code>DecimalField(10,2)</code> | Active offer price in BDT                                                           |
| <code>original_price</code>         | <code>DecimalField(10,2)</code> | Regular list price in BDT                                                           |
| <code>discount</code>               | <code>CharField(50)</code>      | Discount badge string (e.g., <code>'10% OFF'</code> )                               |
| <code>report_time</code>            | <code>CharField(100)</code>     | Formatted report delivery time                                                      |
| <code>home_sample_collection</code> | <code>BooleanField</code>       | Doorstep collection availability                                                    |

## 2.5 Patient Booking Domain

#### DoctorBooking

Records patient chamber appointment bookings.

| Field        | Type           | Constraints          | Description                         |
|--------------|----------------|----------------------|-------------------------------------|
| id           | UUIDField      | PK                   | Booking ID                          |
| user         | ForeignKey     | FK User              | Patient account                     |
| affiliation  | ForeignKey     | FK DoctorAffiliation | Specific doctor-chamber affiliation |
| date         | DateField      | Required             | Appointment date                    |
| slot         | CharField(50)  | Required             | Time slot                           |
| patient_name | CharField(100) | Required             | Patient full name                   |
| status       | CharField(50)  | Default: 'Confirmed' | Booking status                      |

#### LabBooking

Records diagnostic lab test bookings and home sample collections.

| Field         | Type           | Constraints             | Description                          |
|---------------|----------------|-------------------------|--------------------------------------|
| id            | UUIDField      | PK                      | Booking ID                           |
| user          | ForeignKey     | FK User                 | Patient account                      |
| center_test   | ForeignKey     | FK DiagnosticCenterTest | Facility-specific test               |
| pickup_date   | DateField      | Required                | Sample pickup / test date            |
| patient_name  | CharField(100) | Required                | Patient name                         |
| patient_phone | CharField(20)  | Required                | Contact phone                        |
| address       | TextField      | Required                | Delivery / sample collection address |
| status        | CharField(50)  | Default: 'Confirmed'    | Booking status                       |

### 3. Key Architectural Patterns & Junction Logic

#### 3.1 Facility-Agnostic Doctor Affiliation Pattern

A doctor can consult at multiple hospitals and/or diagnostic centers simultaneously. Rather than creating separate tables for Hospital Chambers vs Diagnostic Chambers, `DoctorAffiliation` acts as a unified junction where:

- Exactly one of `hospital` or `diagnostic_center` is populated.
- Custom fee, consultation type ( `OPD` vs `In-patient` ), and schedule slots ( `AffiliationSchedule` ) are attached directly to the affiliation.

#### 3.2 Dual-Facility Test Offering Pattern

Similarly, `DiagnosticCenterTest` serves as a through-model linking tests to either standalone `DiagnosticCenter` units or hospital-internal diagnostic labs ( `Hospital.offered_tests` ). This enables:

- Search filtering across both standalone diagnostic centers and hospital labs.
- Contextual test search keyword filtering inside nested serializer responses.

### 4. Summary of API Endpoints & Filter Parameters

| Endpoint                          | Method | Supported Query Parameters                                                                                    | Description                                               |
|-----------------------------------|--------|---------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| /api/doctors/                     | GET    | specialty , location , area , search ,<br>consultation_type , hospital , diagnostic_center ,<br>fee_max , day | Doctor directory with<br>location & area search           |
| /api/hospitals/                   | GET    | location , area , category , search                                                                           | Hospital directory                                        |
| /api/diagnostic-<br>centers/      | GET    | location , district , area , category , spec , owner ,<br>testcat , search                                    | Diagnostic center search<br>with nested test filtering    |
| /api/tests/                       | GET    | category , search                                                                                             | Master test catalog                                       |
| /api/diagnostic-<br>center-tests/ | GET    | center , hospital , branch , test                                                                             | Price and turnaround time<br>queries                      |
| /api/search-<br>metadata/         | GET    | None                                                                                                          | Unified dropdown<br>metadata (specialties,<br>categories) |
| /api/admin-<br>init/              | GET    | None                                                                                                          | Consolidated initial<br>payload for admin<br>workspace    |